FANFAR SYSTEM OVERVIEW

Jafet Andersson, Senior Researcher, Hydrology Research, SMHI

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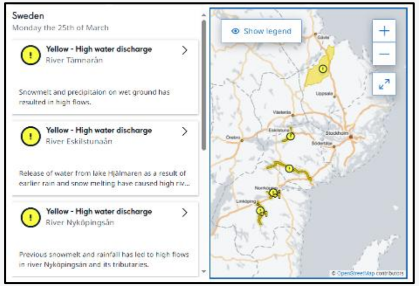
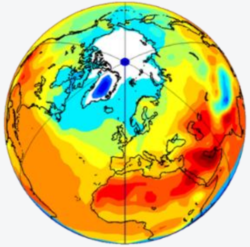
Introduction

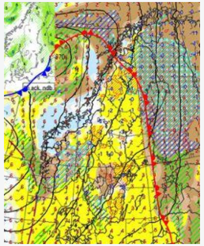
Swedish Meteorological and Hydrological Institute

SMHI

- Swedish government authority
- Weather, water, climate
- Observations, forecasts, warnings, 24/7 operations, climate projections and adaptation, planning support, resource management, infrastructure design
- 600 staff
- Public services, research, business, international cooperation
- Member of WMO, ECMWF, EUMETSAT, EUMETNET etc.

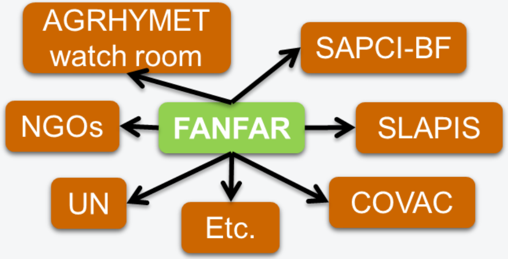






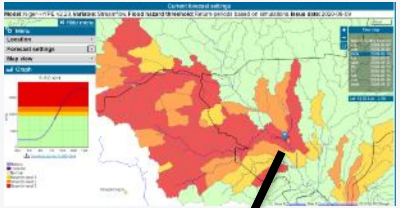
SMHI – West Africa collaboration

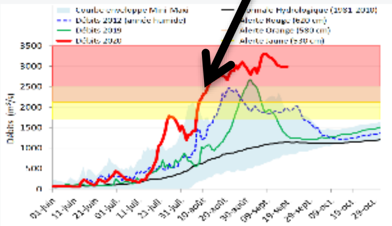
- Series of projects from 2010 to now
- >30 institutions in 17 countries: AGRHYMET RCC-WAS, river basin organisations, national services (meteo. hydro. DRR), partners
- Joint R&D + technical training
- Hydrological modelling, climate impacts, infrastructure design, seasonal forecasts, operational flood forecasting & warning (FANFAR system, since 2018)
- FANFAR becoming regional reference system for hydrological forecasting and flood warning
- https://fanfar.eu/



Co-creation

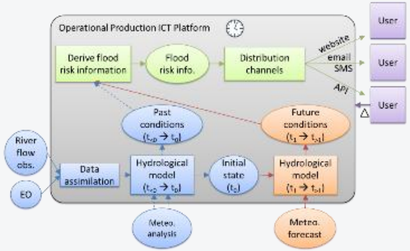
Good forecast of 100-year flood in Niamey, 2020-08-10





AGRHYMET's watch room

Operational forecast production system



2500 lives saved & dam protected in Nigeria (Oct-20)

https://kapital929.fm/flood-about-two-hundred-houses-submerged-by-water-in-jebba-kwara-state/ , used with permission

FANFAR system

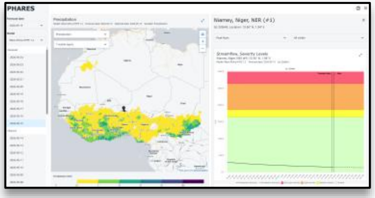
https://fanfar.eu/

FANFAR system on PHARES

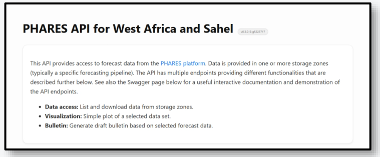
May 2026

24/7 IT production: 4 forecasting chains

Web visualisation portal



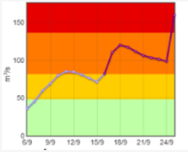
APIs for data access (3)

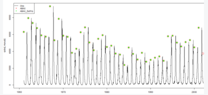


PHARES API for West Africa and Sahel

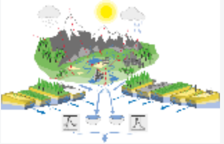
This API provides access to forecast data from the PHARES platform. Data is provided in one or more storage zones (typically a specific forecasting pipeline). The API has multiple endpoints providing different functionalities that are described further below. See also the Swagger page below for a useful interactive documentation and demonstration of the API endpoints.

- **Data access:** List and download data from storage zones.
- **Visualization:** Simple plot of a selected data set.
- **Bulletin:** Generate draft bulletin based on selected forecast data.



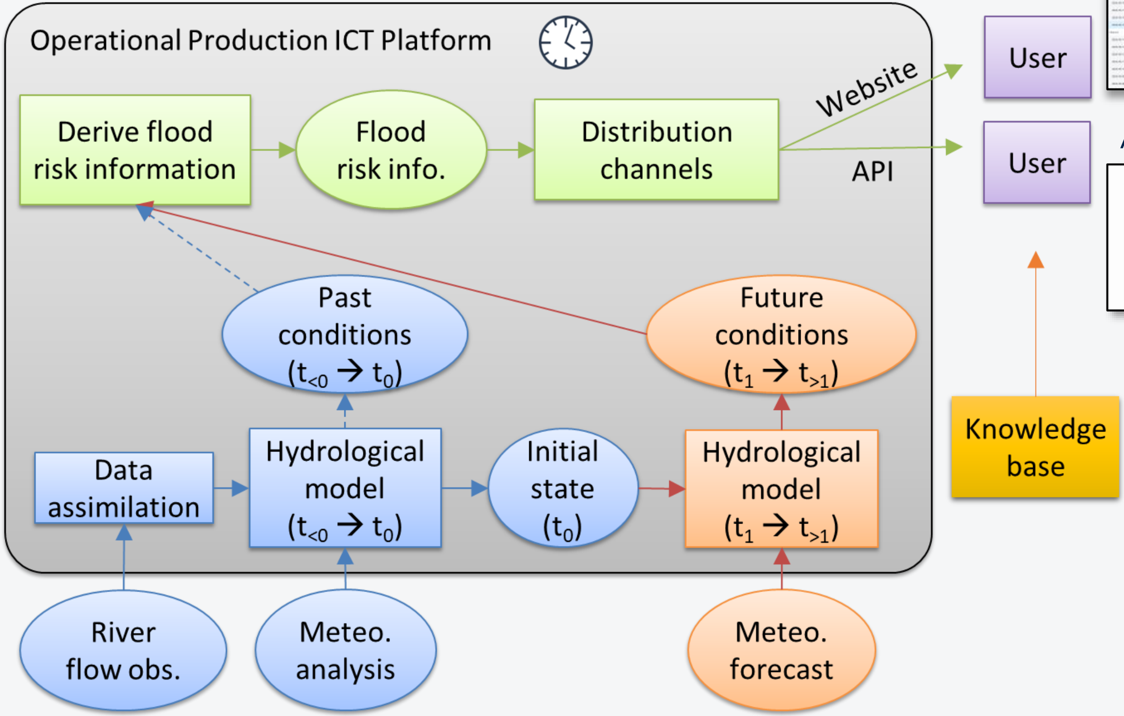


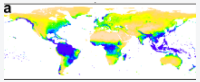
Severity categories (yellow, orange, red) based on peak flow probabilities

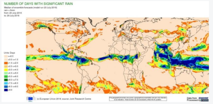


3 hydrological models: West Africa HYPE, Niger-HYPE, Burkina Faso HYPE

Hydrometric stations (Q,WL): a few in progress







2 datasets of past meteorological conditions (1979 to yesterday): HydroGFD, CHIRPS

3 datasets of forecasted meteorological conditions (today and future): ECMWF, CHIRPS-GEFS, WRF-ANAM

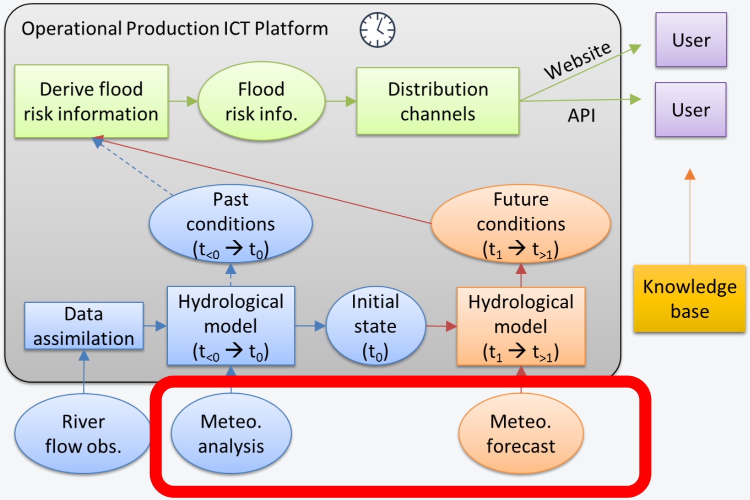
Blue: initialisation / analysis
Orange: forecast
Green: post-processing
Grey: IT system

CILSS

SMHI

FANFAR

Sweden Sverige



Meteorological input data

General aspects about meteorological input data

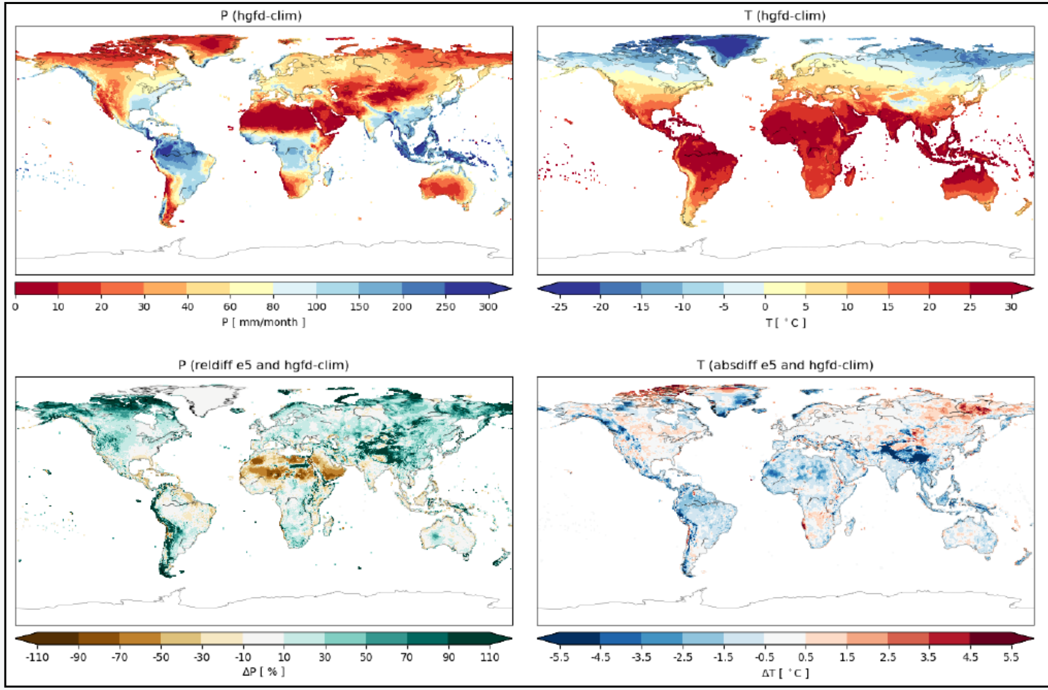
- Essential forcing data for hydrological models
- Daily (or hourly) temporal resolution
- Complete spatial and temporal coverage required (grids)
- Essential variables: precipitation, air temperature (min, mean, max)
- Historic conditions used for model calibration, and model initialisation (e.g. 1979-2020)
- Recent and future conditions used for forecasting
- Requires combination of several datasets
- To avoid biases, calibration data should be used in operation as much as possible, and additional (forecast) data should have similar characteristics (e.g. bias-adjusted toward historic calibration data)

Meteorological data chain 1: HydroGFD3 + ECMWF operational deterministic forecasts

HydroGFD version 3.2 (past conditions)

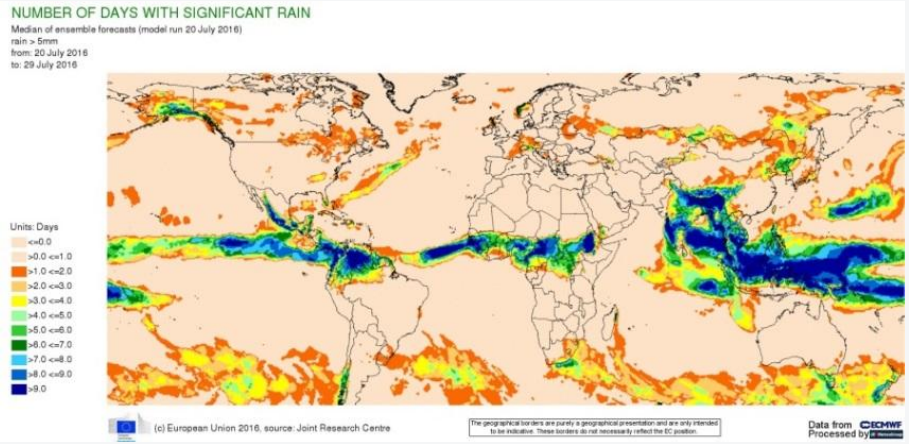
- Produced by SMHI
- Global, 0.25° grid of precipitation and temperature (mean, min, max)
- 1940 to [today - 5 days], daily & hourly
- ERA5 reanalysis + bias-correction against reference gauge and satellite data (GPCC, CRU, CPC)
- Correction 1 (P): number of wet days
- Correction 2 (P&T): monthly values scaled with ratio between ERA5 and reference
- https://doi.org/10.5194/essd-13-1531-2021 (article & open data)

Climatology 1980-2009 of HydroGFD vs ERA5. Bias of ERA5 is substantial in many places



ECMWF operational deterministic meteorological forecasts (future)

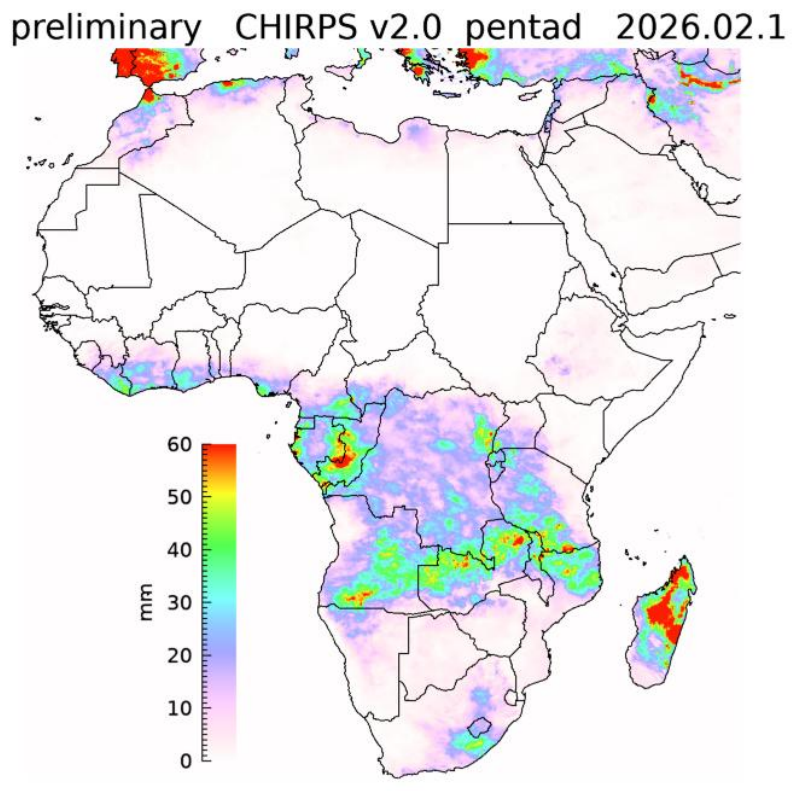
- Operational deterministic meteorological forecast, 1-10 days forward in time
- Global, 0.125° grid (future: 0.1°)
- IFS versions change regularly
- Tailored by SMHI to HydroGFD3: daily resolution and scaled to same grid (0.25°)
- First day from 5 past forecasts (OD) used to fill gap between latest HydroGFD and current forecast (ODF)
- https://www.ecmwf.int/en/forecasts/datasets/open-data



Meteorological data chain 2: CHIRPS + HydroGFD + CHIRPS-GEFS + ECMWF OD

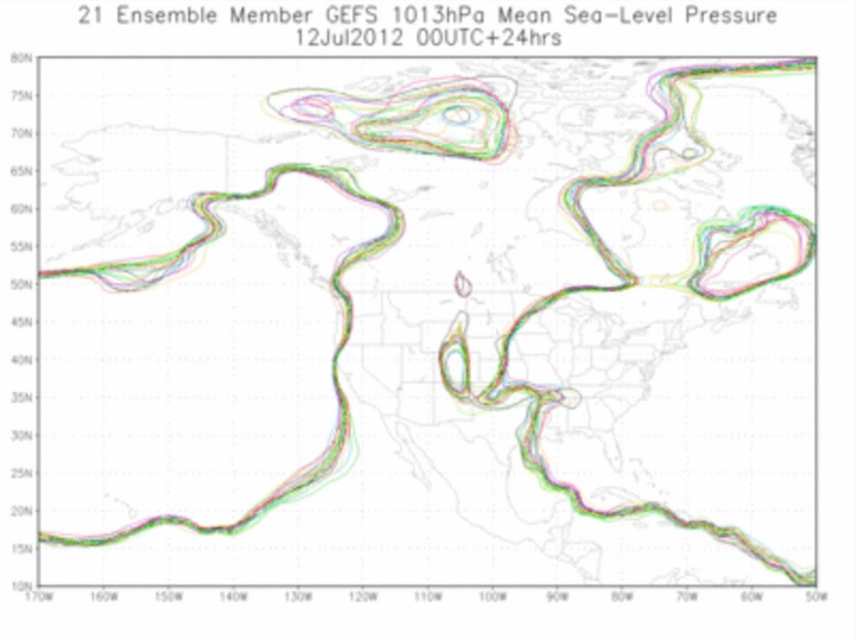
CHIRPS & CHIRP (past conditions)

- Precipitation only
- Produced by UCSB, open data
- CHIRPS = Climate Hazards Center InfraRed Precipitation with Station data
  - Climatology + satellite + gauges
  - 0.05° grid, quasi-global
  - 1981 to [today – 2months]
  - Version 2 used now, v3 available
- CHIRP:
  - same but without gauges
  - available: [today – 4 to 10 days]
- https://chc.ucsb.edu/data



CHIRPS-GEFS meteorological forecasts (future conditions)

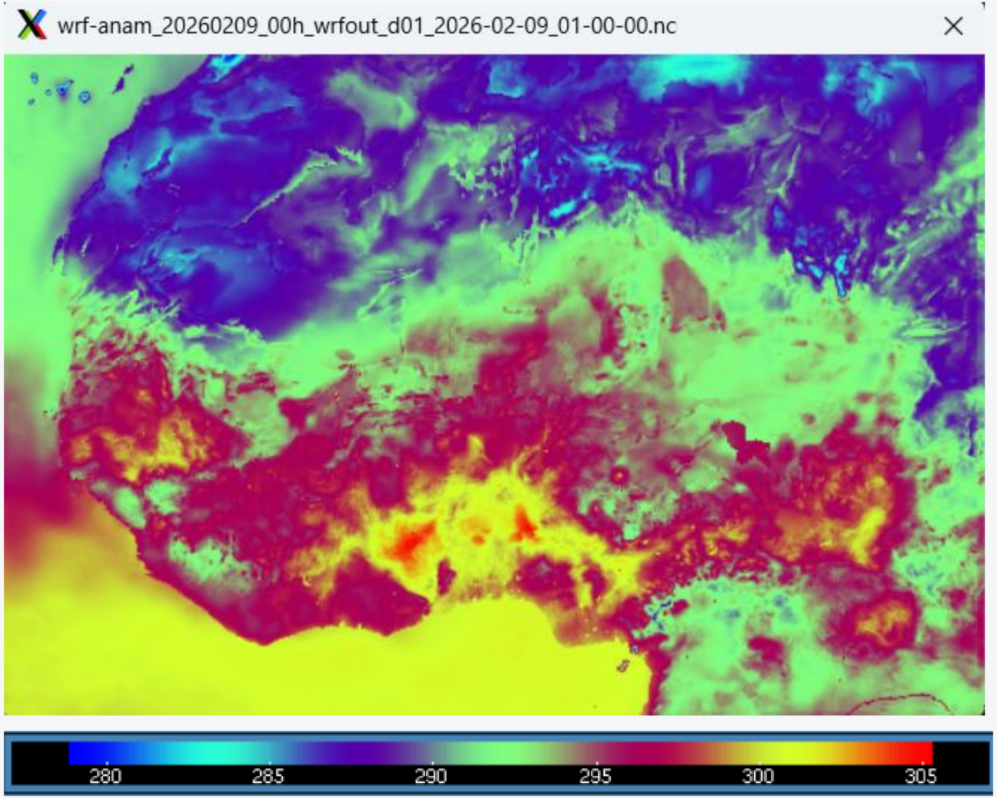
- NOAA global ensemble forecast system (GEFS), 1-16 days forward in time
- Tailored by UCSB to CHIRPS: bias-adjustment and scaled to same grid (0.05°)
- First day from past forecasts used to fill gap between latest CHIRP time step and current forecast
- Currently using data based on CHIRPS v2, but v3 is available
- https://chc.ucsb.edu/data



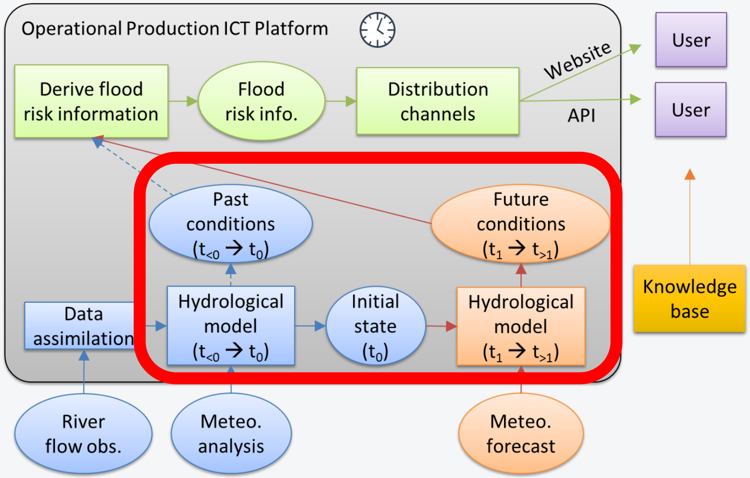
Meteorological data chain 3: CHIRPS + HydroGFD + WRF-ANAM

WRF from ANAM in Burkina Faso (future conditions)

- WRF model set up for West Africa domain (~0.08°)
- No tailoring to CHIRPS or HydroGFD
- First day from past forecasts used to fill gap between HydroGFD/CHIRPS and current forecast
- Downloaded from ANAM FTP



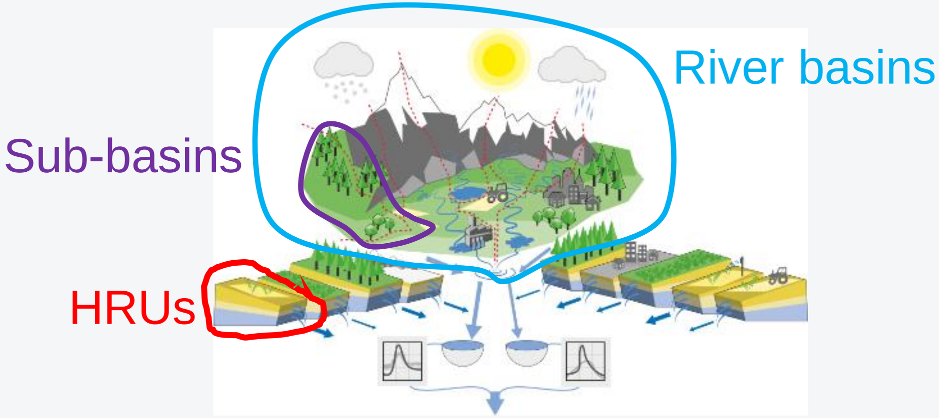
Example: air temperature 2026-02-09 01:00 (° K)



Hydrological models

HYPE hydrological model core

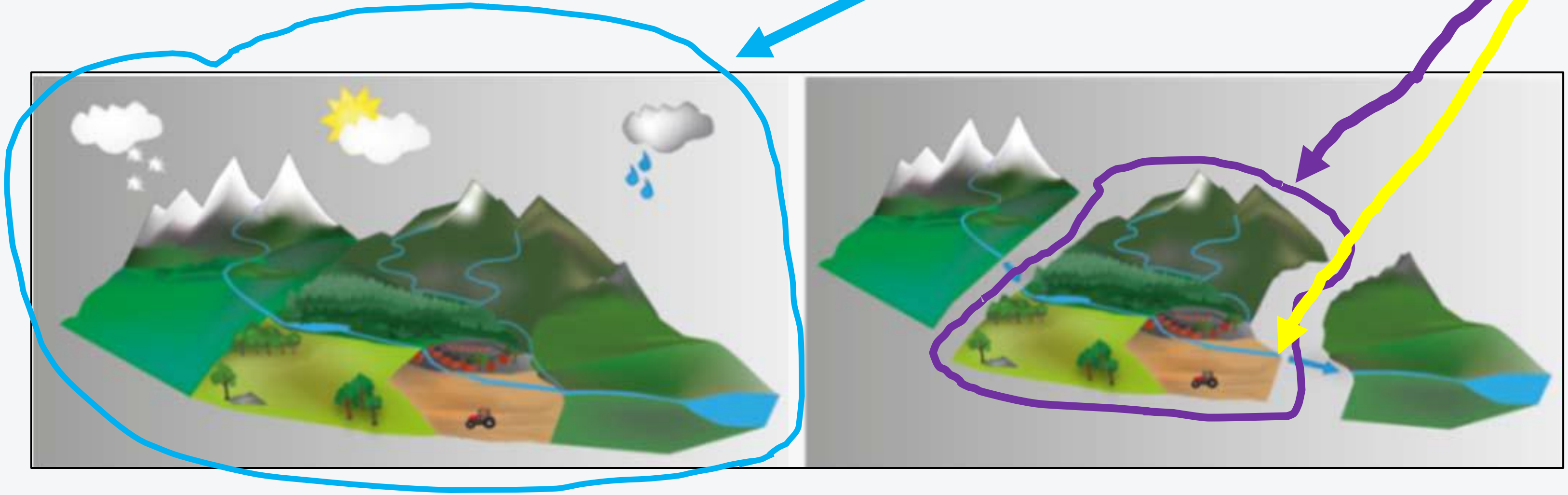
- Natural and man-made processes, focus on major dynamics and large scales (precipitation → streamflow)
- Semi-distributed
  - Topography data gives sub-basins and river network
  - HRU fractions in sub-basins are unique soil and land use combinations
- Requires meteorological forcing data (P, T etc.)
- Calibrated against discharge observations (and sometimes ET, snow etc.)
- Typical subbasin resolution: 1-2000km^2
- Temporal resolution: daily, hourly
- Can assimilate real-time observations (e.g. discharge)



- Developed by SMHI, used operationally in Sweden and many other countries
- Free and open source : https://hypeweb.smhi.se/model-water/
- Free course annually: https://hypeweb.smhi.se/model-water/courses-and-training/
- Wiki: http://hype.smhi.net/wiki/

HYPE variables

- Water fluxes (e.g. discharge) and water stores (e.g. soil moisture)
- Different types of variables, e.g.
  - discharge (q, *cout*) is provided at the sub-basin outlet
  - actual evapotranspiration (*evap*) is for subbasin area
  - upstream precipitation (*upcprc*) is for entire upstream catchment area

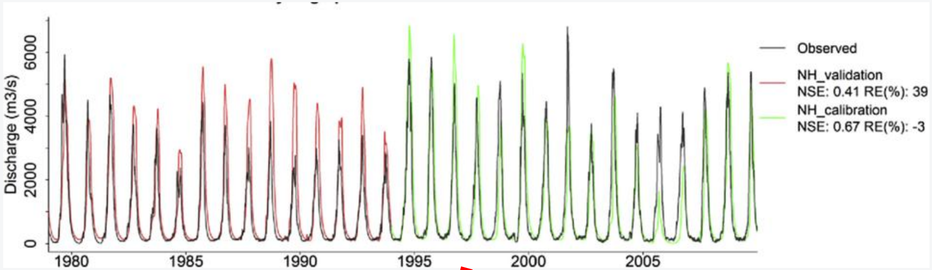


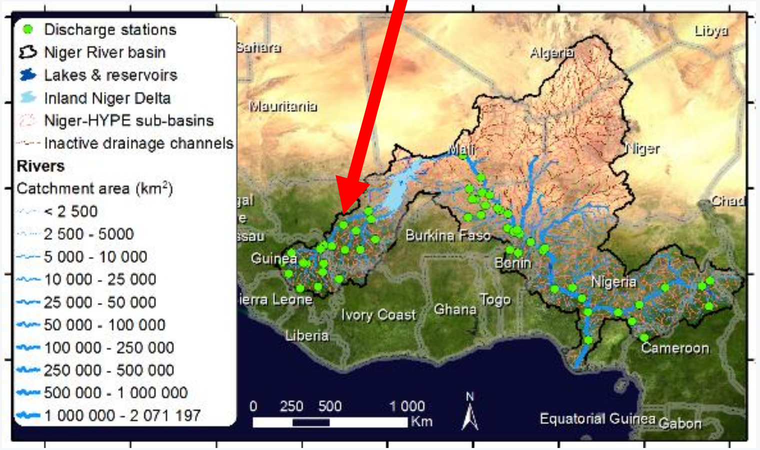
Current HYPE models in FANFAR

Niger-HYPE version 2.30 (*niger_hype*)

- Domain: Niger River basin
- ~800 subbasins with average resolution ~2600 km^2
- Jointly developed by SMHI & AGRHYMET in 2017(+)
- Process refinements particularly focussed on Inland Niger Delta (IND) and surface vs. sub-surface runoff flowpaths
- Calibrated toward 56 discharge stations and satellite PET
- Meteo calibration data: WFDEI (similar to HydroGFD)
- Performance: median daily NSE = 0.4 (range -16 to 0.89)
- https://doi.org/10.1002/hyp.11376

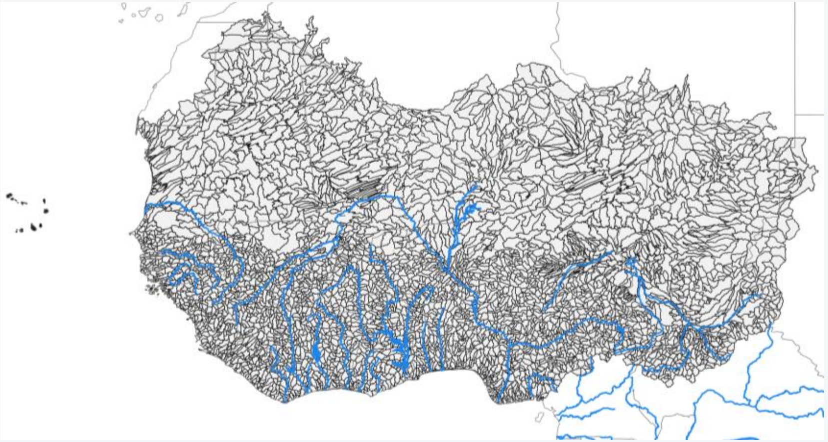
Koulikoro, Mali – sim vs. obs

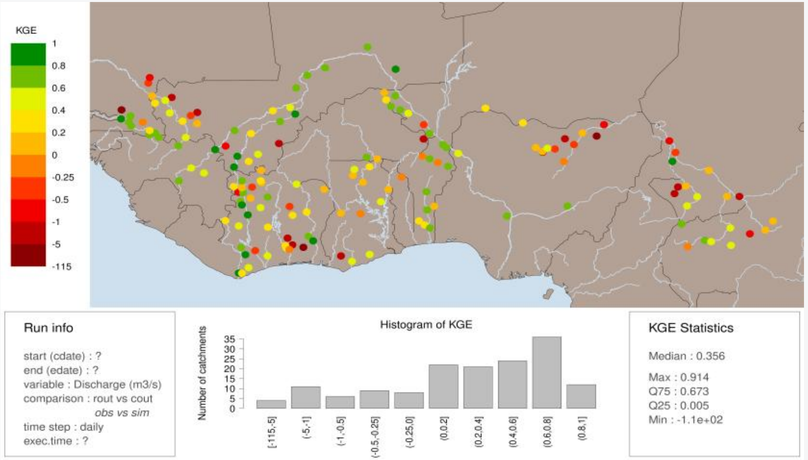




West Africa HYPE version 1.2 (*wa_hype*)

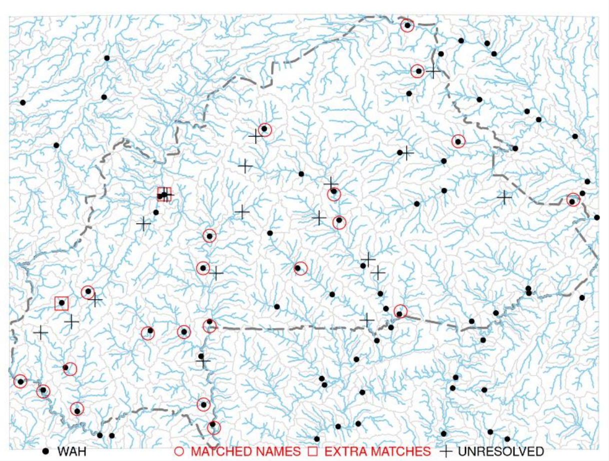
- Domain: West Africa
- ~4600 subbasins with average resolution ~1000 km^2
- Based on global World-Wide HYPE model v1.3 ([Arheimer et al 2020]())
- Adapted to West Africa: catchment delineation, calibration of runoff generation, water balance, lakes and IND using ~150 discharge stations
- Meteo calibration data: HydroGFD3
- Performance: median daily KGE = 0.36 (range -115 to 0.91)
- Description of [WA HYPE]() (section 4)



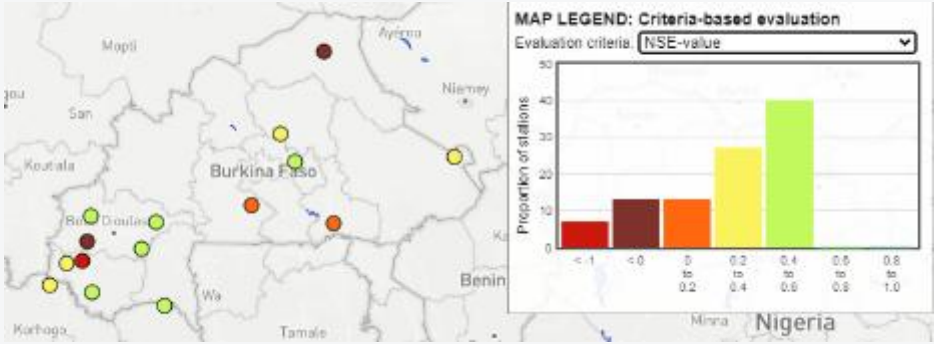


Burkina Faso HYPE 1.1 (*bf_hype*)

- Domain: Burkina Faso (+ upstream catchments)
- Built on Africa-HYPE (precursor to coming World-Wide HYPE v.2 with more soils)
- ~300 subbasins with average resolution ~1000 km^2
- Calibrated for Burkina Faso with 15 discharge stations from DEIE/DGRE (matching location and upstream area, >5 years data), focus on soils, evaporation and reservoirs
- Meteo. calibration data: CHIRPS2 (slightly better dynamics than HydroGFD3)
- Performance (v1.0): median daily NSE = 0.34 (range -0.2 to 0.5)

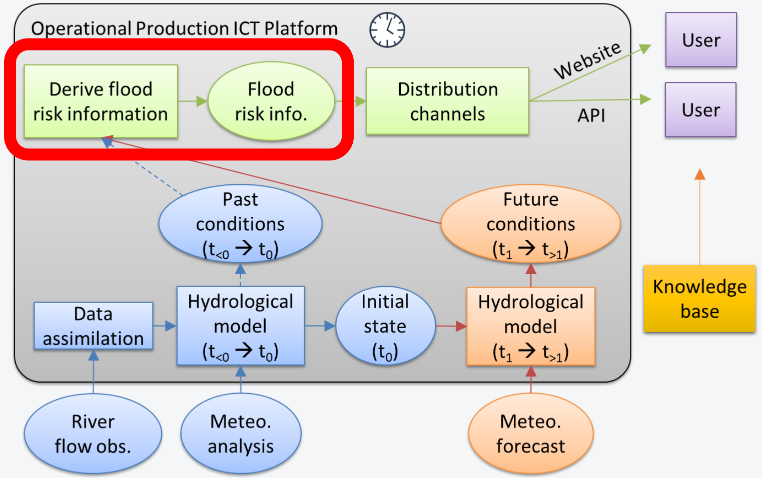


NSE for BF-HYPE 1.0



Current forecasting chains

1. West Africa HYPE 1.2 + HydroGFD 3.2 + ECMWF OD
2. Niger HYPE 2.30 + HydroGFD 3.2 + ECMWF OD
3. Burkina Faso HYPE 1.1 + CHIRPS2 (P) & HydroGFD 3.2 (T) + CHIRPS-GEFS
4. Burkina Faso HYPE 1.1 + CHIRPS2 (P) & HydroGFD 3.2 (T) + WRF-ANAM



Determining flood hazard

Is current river flow a flood hazard?

SMHI

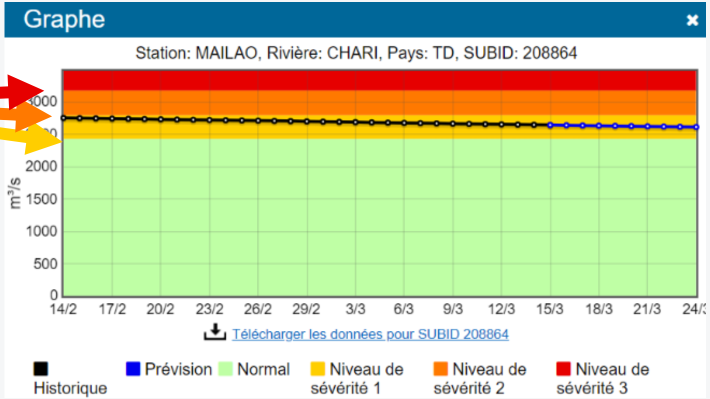
Current approach

- Discharge + threshold = severity level of flood hazard. Calculated for each sub-basin and each day
- Thresholds are based on frequency of simulated peak flows

Possible future approaches

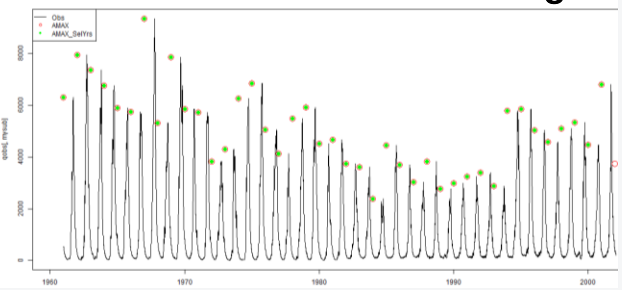
- Local thresholds
- % of a selected historic year
- Water levels (requires rating curves)

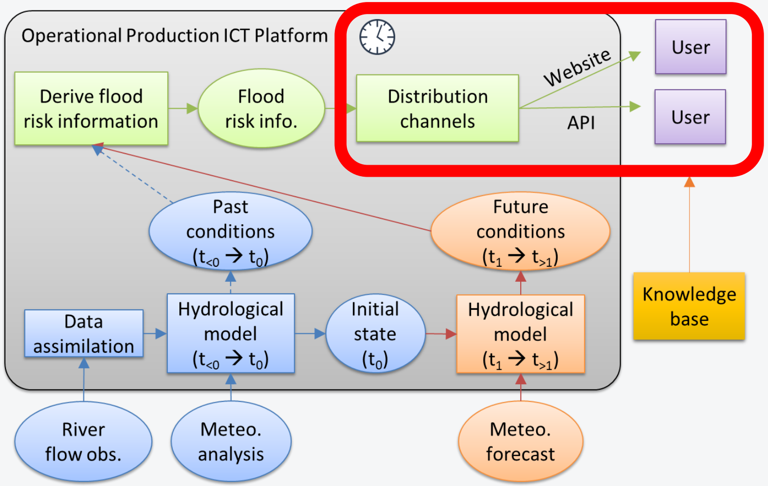
Frequency analysis (GEV)



Return period	Alert level
< 2-year	0 (green)
> 2-year	1 (yellow)
> 5-year	2 (orange)
> 30-year	3 (red)
* When GEV model fails *	-99 (grey)

Annual maximum discharge



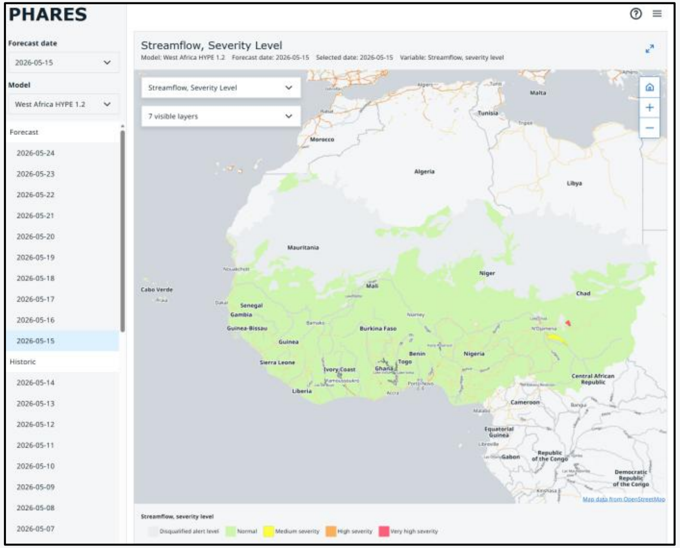


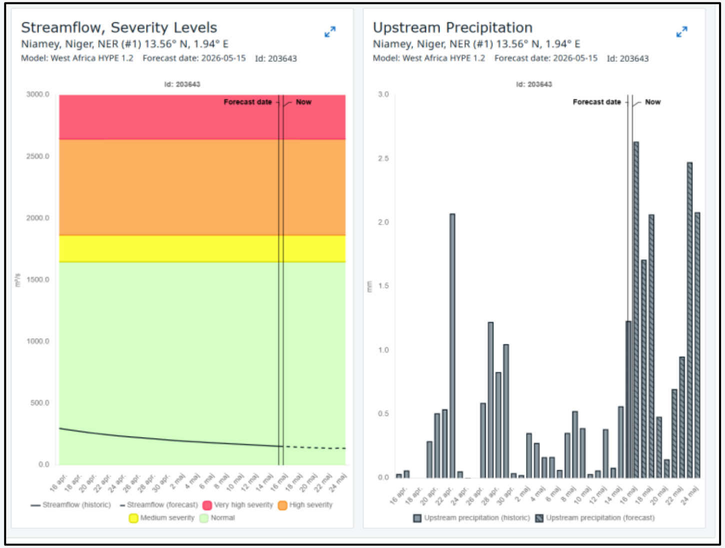
Access to forecast information

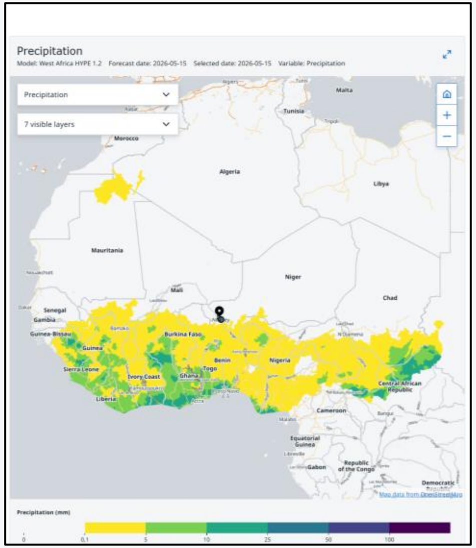
Visualisation platform

SMHI

- Multiple models (forecast chains)
- Variables: flood hazard severity levels, streamflow, precipitation (*more is coming*)
- Current & past forecasts
- Maps of all catchments
- Graphs for specific catchment
- Bookmarks (e.g. hydrometric stations)
- Standard inputs formats
- Built with HyVis tool







APIs

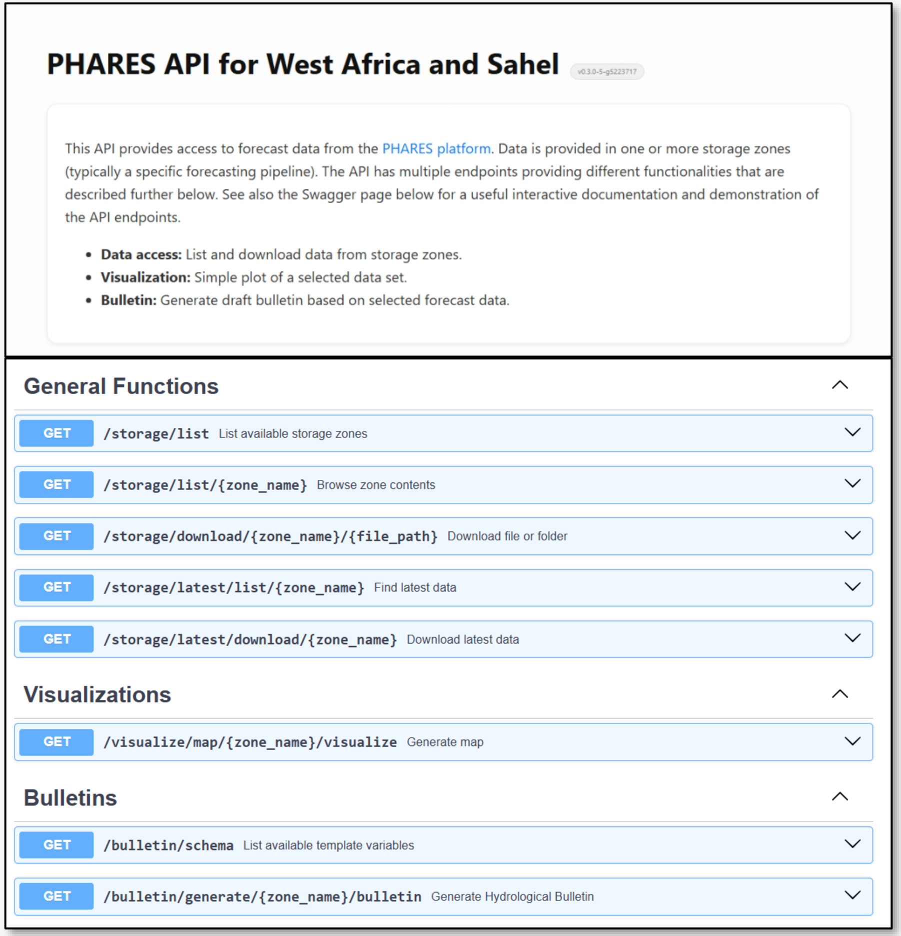
- Built with FastAPI framework

Key functions

- Data access: list and download data for forecasting chains
- Visualisation: simple static map of a selected data
- Bulletin: Generate draft bulletin based on selected forecast data

Multiple APIs

- WAS: West Africa and Sahel
- BFA: Burkina Faso
- demo: example data
- *more coming*



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- **Data access:** List and download data from storage zones.
- **Visualization:** Simple plot of a selected data set.
- **Bulletin:** Generate draft bulletin based on selected forecast data.

General Functions

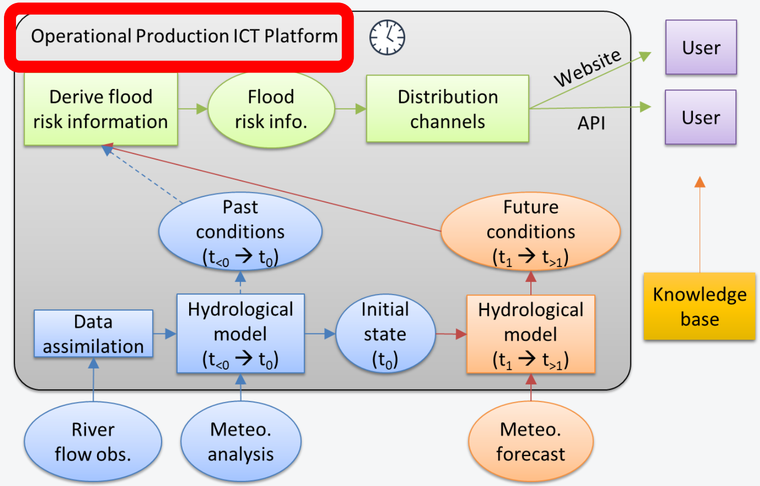
- GET `/storage/list` List available storage zones
- GET `/storage/list/{zone_name}` Browse zone contents
- GET `/storage/download/{zone_name}/{file_path}` Download file or folder
- GET `/storage/latest/list/{zone_name}` Find latest data
- GET `/storage/latest/download/{zone_name}` Download latest data

Visualizations

- GET `/visualize/map/{zone_name}/visualize` Generate map

Bulletins

- GET `/bulletin/schema` List available template variables
- GET `/bulletin/generate/{zone_name}/bulletin` Generate Hydrological Bulletin

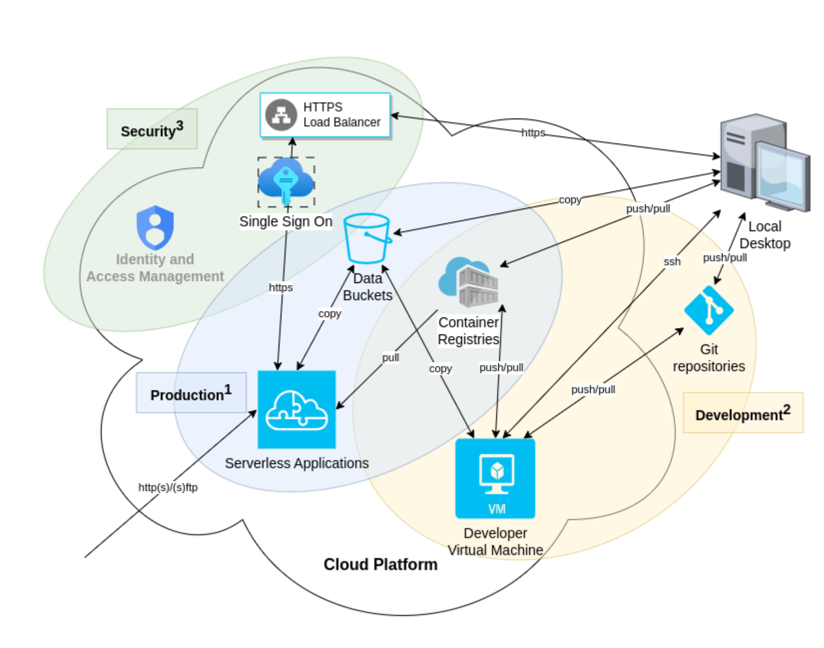


IT environment for operational production

IT environment for operational production

- FANFAR is first system implemented on PHARES, which is AGRHYMET's new multi-model cloud-based platform for operational forecasting of hydro-climatic extremes
- PHARES is implemented on Google Cloud Platform (GCP)
- Why cloud: robust, scalable, affordable, less IT expertise needed → more time to focus on the topic (hydrology / meteorology / warnings)

IT architecture and key GCP components



SMHI

Merci!
Thank you!

FANFAR

www.fanfar.eu